

Find out all the list of companies which are conducting Placement drive using dsa?

Global Tech Giants:

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1. Google

9. Oracle

2. Microsoft

10. Salesforce

3. Amazon

11. Uber

4. Meta

12. Cisco

5. Apple

E-commerce & product-Based Companies

6. Adobe

Financial & Banking Tech

7. Flipkart

13. Goldman Sachs

8. LinkedIn

14. Morgan Stanley

(IT Services) Mass Recruiters

15. TCS

19. IBM

16. Infosys

20. Deloitte

17. Wipro

21. Accenture

18. Cognizant

2) What is the syllabus of dsa in Gate?

a) Programming and Data Structures

→ Programming in C: Basics, data types, control structures, functions, pointers, arrays, structures, and unions

→ Recursion: Concepts, implementation & application

→ Data structures: Arrays, stacks, queues, linked lists, trees, binary search trees, binary heaps & graphs

b) Algorithms

→ Searching & Sorting: Linear search, binary search, hashing, & basic sorting algorithms

→ Complexity: Asymptotic worst-case time & space complexity analysis

→ Design Techniques: Divide-and-conquer, Greedy approaches, & Dynamic Programming

→ Graph Algorithms: Traversals (BFS, DFS), minimum Spanning trees & shortest paths (Dijkstra, Floyd-Warshall)

3) what are the set of variables in amazon invoice bill

```
String Billing_Address;  
int UTcode; String UTcode;  
String Shipping_Address;  
String State;  
String place_of_supply;  
String place_of_delivery;  
String Invoice-Number;  
String Invoice-Details;  
String Invoice-Date;  
String Seller-Name;  
String seller-Address;  
String PAN-NO;  
String gst-Registration No;  
String Order-Number;  
String Order-Date;  
String description-of-item;  
double Unitprice;  
int Quantity;  
double Net-Amount;  
double Tax-Rate;  
String Tax-type;  
double Tax-Amount;  
double Total-Amount;  
String Ammount-in-words;
```

4) Gas bill

```
String consumer_Name;  
unsigned int consumerNo;  
String LPDID;  
String Address;  
unsigned int mobileno;  
String Category;  
String product/HSN;  
String connection;  
double quantity;  
String ctc(status);  
double subsidy;  
double consumed;  
String Hose-Expiry;  
String MI;  
String E-KYC;
```



```

string BookingSource;
string payment;
string orderDate;
unsigned int orderNo;
unsigned int cashmemoNo;
unsigned int
string double cashmemoDate;
double baseprice;
double deliverycharges;
double c&cRebate;
double taxableAmount;
double cgst;
double sgst;
double totalAmount;

```

5) Electricity Bill

```

String Bill-date;
string Time;
unsigned int Bno;
String ERO;
String SEC;
String DIST;
String SCNO;
unsigned int uniqueScno;
String name;
String Address;
String SCAT;
String Cat;
int CAT;
int PH;
string double contractedLoad;
double connectedLoad;
int mno;
int double MF;
string double netbill;
double netbill;
String dueDate;
String disDate;
unsigned int phoneno;
unsigned int dtrcode;
unsigned int frcode;
String Reading;
String month;
double Charges;
double energyCharges;
double electricityDuty;
double cgst;
double sgst;
double totalAmount;

```

→ Define a structure in C & C++-This structure should hold the ~~data~~ ^{type} variable for student, compute the size of structure manually. Use the size of function & display the size. cross check if the both are same or not. Justify the answer.

Structure in c: A structure is a user-defined data type that allows you to group together variables of different data types under a single name. It is purely a data container & can only hold variables (members) not functions.

Syntax:

```
struct student {  
    int rollno;  
    float marks; };
```

```
struct student s1;
```

Structure in c++: A structure is a user-defined datatype that can hold both variable & member functions (including constructors & destructors). It behaves almost identically to a class, but its members are public by default.

Syntax:

```
struct student {
```

```
    int rollno;
```

```
    float marks;
```

```
    void display() {
```

```
    } };
```

```
student s1; // no need of struct keyword
```

→ Differences

Feature	structures in c	Structure in c++
Functions inside	Not Allowed	Allowed
Access Specifiers	Not supported	Supported (public, private, protected)

keyword needed for object? Yes No
supports inheritance No Yes
size of empty struct '0' Bytes 1 Byte
(or compiler error)

// WAP on structure that related to structure
// to display the size of structure, which containing
// the variables

```
#include <stdio.h>
struct student {
    int roll; // → 4
    char name[15]; // → 15
    double marks; // → 8
}; // 4+15+8 = 27
```

```
int main() {
    struct student s;
    printf("The size of structure is: %ld", sizeof(s));
    return 0;
}
```

Expected output: 27

Actual output: 32.

The difference b/w expected output & actual output is due to ; here int roll occupies 4 bytes (bytes 0-3), char name[15] occupies 15 bytes (bytes 4-18), up to here 19 bytes are used, because the next member (double) memory address that is a multiple of 8, the compiler can't place it at byte 19.

- The compiler automatically inserts 5 bytes of empty padding (bytes 19-23) to jump to the next 8 byte (byte 24-31). This is called padding.

- Find out the applications of recursion.
- (1) is recursion still used in present programming era? If so where?
- what are programming languages that
- (2) Supports recursion calls?

1) Application of recursion :

1. Tree and Graph Traversal: Used for systematic-ally exploring nodes / vertices in data structures like trees & graphs.
2. Sorting Algorithms: Algorithms like quicksort and merge sort divide data into subarrays, sort them recursively, and merge them.
3. Divide-and-conquer Algorithms: Algorithms like binary search break problems into smaller using recursion.
4. Fractal Generation- Recursion helps generate fractal patterns, such as the mandelbrot set, by repeatedly applying a recursive formula.
5. Backtracking Algorithms: Used for problems requiring a sequence of decisions, where recursion explores all possible paths and backtracks when needed.
6. Memoization: Involves caching results of recursive function calls to avoid recomputing expensive subproblems.
7. Dynamic Programming: Recursive solutions with memoization are used in problems like Fibonacci, knapsack, & longest common subsequence.
8. Mathematical Computations: Calculating factorials, fibonacci numbers, greatest common divisor (GCD), & power.

Yes, recursion is absolutely still used in the present programming era. It is especially useful for problems that can be broken into smaller versions of the same problem. Recursion remains important in many areas where it makes the code simpler and more natural.

Where is it Used Today?

- web browsers: parsing HTML & the document object model
- Operating systems: Navigating directories & file system.
- Compilers: Parsing source code & syntax trees.
- Artificial Intelligence: Search algorithms & game trees.
- Databases: Recursive queries (such as SQL Common Table Expressions).
- Machine learning & Data structure.
Tree-based algorithms like decision trees.

Recursion is still an essential programming technique in the present era. It is widely used in algorithms, data structures, AI, Compilers, O.S, & web development. While iteration is preferred for simple loops,

recursion is often the best choice for solving hierarchical & divide-and-conquer problems.

2) many modern programming languages support recursion. Here are some of most common ones.

- | | | |
|-----------|----------------|-------------------|
| 1. C | 6. Javascript | 11. Scala |
| 2. C++ | 7. Go (Golang) | 12. Haskell |
| 3. Java | 8. Rust | 13. LISP / Scheme |
| 4. python | 9. Swift | 14. PHP |
| 5. C# | 10. Kotlin | 15. Ruby |

Note: while most programming languages support recursion, each language has a maximum recursion depth or stack size. Deep recursion can cause a stack overflow, so for very large problems, an iterative approach or optimized techniques (such as tail recursion, where supported) may be preferred.

- H.W
1. → Find out the History of towers of hanoi.
 2. → Today from the morning 7:30 A.M to 4:30 PM
 3. → List out the various dsa we observed in your surroundings.

1- The towers of hanoi was invented in 1883 by the French mathematician Edouard Lucas. If his name sounds familiar to math enthusiasts, it's because he is also famous for studying the Fibonacci Sequence and discovering the Lucas numbers (1, 3, 4, 7, 11, ...).

Lucas published the under an alias, v. claus de siam (an anagram of Lucas d'Amiens, Amiens being his birthplace), claiming it came from the far east to give it an exotic, mysterious allure.

The legend of Benares (The Myth) : To make the puzzle sell, an elaborate, apocalyptic backstory

accompanied its release. According to marketing material provided with the toy:

In the great temple of Benares (Varanasi, India), beneath the dome which marks the center of the world, rests a brass plate in which are fixed three diamond needles, each a cubit high & as thick as the body of a bee.

On one of these needles, God placed 64 discs of pure gold at the creation, the largest disc resting on the brass plate, & the others getting smaller and smaller up to the top one.

This is the Tower of Brahma.

Day & night, the priest transfer the discs from one diamond needle to another according to the immutable laws of Brahma:

1. You can only move one disc at a time.
2. A larger disc can never be placed on top of a smaller one.

The End of the world:

The legend concludes with a dramatic prophecy: The moment the priests successfully moves all 64 discs to the third needle, the temple will crumble into dust, and the world will vanish with a thunderclap.

The Mathematical Reality: The math proves the world is safe for a very long

The minimum number of moves required to solve a Tower of Hanoi puzzle with n discs is given by the formulae:

$$\text{Moves} = 2^n - 1$$

For the priests' puzzle with 64 discs, the calculation is:

$$2^{64} - 1 = 18,446,744,073,709,551,615 \text{ moves}$$

If the priests were incredibly fast & moved one disc every single second without making a single mistake, it would take them

• Over 584 billion years to complete the task. Given that our universe is only about 13.8 billion years old, the world isn't ending anytime soon.

Why it matters Today: while it started as a wooden toy in 19th century France, The tower of Hanoi became incredibly important in the 20th & 21st centuries for computer Science.

Because the puzzle can be broken down into smaller version of itself (to move 64 discs, you must figure out how to move 63 discs, which requires moving 62 & so on), it is the gold standard for teaching recursion & algorithm design to programming students worldwide.

2. I observed
- i) choosing the shortest route from home to college - graph
 - cii) Bus seating arrangement - array
 - ciii) people getting out of the bus one by one - Queue
 - civ) waiting to board the bus - Queue
 - cv) The books in a bag - Stack
 - cvi) Searching for a contact in a mobile phone - Linear Search
 - cvi) Attendance checked one student after another - searching
 - cvi) class timetable arranged by time - Sorting
 - cix) Road network connecting different places - Graph
 - x) The history in a google search - Stack.